

COURSE TITLE : **ADVANCED WEB DESIGN**
 COURSE CODE : **5262**
 COURSE CATEGORY : **A**
 PERIODS/WEEK : **4**
 PERIODS/SEMESTER : **52**
 CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	HTML Document Object Model (DOM) and javaScript Object Notation (JSON)	13
2	eXtensible Mark-up Language (XML)	13
3	Asynchronous JavaScript And XML (AJAX)	13
4	jQuery	13

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To understand HTML Document Object Model (DOM) and javaScript Object Notation (JSON)
2	1	To understand eXtensible Mark-up Language (XML)
3	1	To understand Asynchronous JavaScript And XML (AJAX)
4	1	To understand jQuery

Specific Outcomes:

MODULE – I: HTML Document Object Model (DOM) and javaScript Object Notation (JSON)

1.1 Understand HTML Document Object Model (DOM) and JSON

- 1.1.1 Define HTML Document Object Model (DOM)
- 1.1.2 Describe Parsing HTML documents and DOM and Tree generation
- 1.1.3 Describe how to manage HTML DOM with javaScript
- 1.1.4 List javaScript inbuilt objects
- 1.1.5 Describe document object

- 1.1.6 Describe window object and its functions, screen object and its functions
- 1.1.7 Explain Event binding methods in javaScript
- 1.1.8 Explain multiple element binding
- 1.1.9 List DOM Element selectors
- 1.1.10 Use javaScript arrays, javaScript objects, array of objects and for-in loop
- 1.1.11 Summarize javaScript Object Notation (JSON)
- 1.1.12 Explain how to parse JSON

MODULE – II: eXtensible Mark-up Language (XML)

2.1 Understand eXtensible Mark-up Language (XML)

- 2.1.1 List javaScript Active objects
- 2.1.2 Describe Time management in javaScript
- 2.1.3 Describe Time object and its functions
- 2.1.4 Explain how to manage System time and UTC
- 2.1.5 Use JavaScript timer objects
- 2.1.6 Describe Time Interval management and Timeout Management
- 2.1.7 Define eXtensible Markup Language (XML)
- 2.1.8 Discuss need for XML
- 2.1.9 Describe XML syntax and structure
- 2.1.10 List applications of XML in Technologies
- 2.1.11 Describe XML tags
- 2.1.12 Explain XML nodes and attributes
- 2.1.13 Define XML DOM
- 2.1.14 Write XML DOM management using JavaScript
- 2.1.15 Show how to get data from XML string

MODULE – III: Asynchronous javaScript And XML (AJAX)

3.1 Understand Asynchronous javaScript And XML (AJAX)

- 3.1.1 Define Asynchronous javaScript And XML (AJAX)
- 3.1.2 List the need for AJAX
- 3.1.3 Discuss the History and Evolution of AJAX
- 3.1.4 List the Application Areas of AJAX
- 3.1.5 Describe The XMLHttpRequest Object and Protocol
- 3.1.6 Differentiate HTTP and XMLHttpRequest
- 3.1.7 Discuss how to create XMLHttpRequest object in javaScript
- 3.1.8 Explain generating and customizing requests with XMLHttpRequest object

- 3.1.9 Describereadystatechange event in XMLHttpRequest object
- 3.1.10 Explain sending requests to server in synchronous & asynchronous methods
- 3.1.11 Distinguish sending requests in GET and POST methods
- 3.1.12 Discuss getting response from server using XMLHttpRequest
- 3.1.13 Distinguish dynamic and static responses
- 3.1.14 Distinguish managing text response and XML response

MODULE – IV: To understand jQuery

4.1 Understand jQuery

- 4.1.1 Define jQuery
- 4.1.2 List need for jQuery
- 4.1.3 Discuss problems with JavaScript programming
- 4.1.4 List advantages of using jQuery
- 4.1.5 Explain jQuery syntax
- 4.1.6 Show how to integrate jQuery into a webpage
- 4.1.7 Define ready function
- 4.1.8 List element selectors
- 4.1.9 Distinguish events and binding method
- 4.1.10 Discuss HTML DOM Management
- 4.1.11 Show accessing element properties and changing element properties
- 4.1.12 Show managing element styles and classes and element dimensions
- 4.1.13 Explain HTML DOM traversal
- 4.1.14 Describe Selector filtering
- 4.1.15 Use jQuery effects like hide, slide, fade, toggle, stop
- 4.1.16 Use jQuery animations
- 4.1.17 Distinguish call-back functions and timer functions
- 4.1.18 Discuss AJAX using jQuery
- 4.1.19 Distinguish LOAD, GET and POST functions
- 4.1.20 Describe jQuery form validation
- 4.1.21 Use jQuery multi-level navigation menu

CONTENT DETAILS

MODULE – I: HTML Document Object Model (DOM) and javaScript Object Notation (JSON)

Introduction to HTML Document Object Model – Parsing HTML documents and DOM Tree generation - Managing HTML DOM with javaScript – javascript inbuilt objects - document object – window object and it's functions – screen object and it's functions - Event binding methods in javaScript – multiple element binding - Dom Element selectors – javaScript arrays – javaScript objects – array of objects – for-in loop – javaScript Object Notation(JSON) – parsing JSON.

MODULE – II: eXtensible Mark-up Language (XML)

Introduction to javaScript Active objects – Time management in javaScript- Time object and it's functions – Managing System time and UTC – JavaScript timer objects – Time Interval management – Timeout Management - Introduction to eXtensibleMarkup Language(XML) – need for XML - XML syntax and structure – applications of XML in Technologies – XML tags - XML nodes and attributes – XML DOM – XML DOM management using JavaScript – getting data from XML string.

MODULE – III: Asynchronous javaScript And XML (AJAX)

Introduction to Asynchronous javaScript And XML (AJAX) – Need for AJAX – History and Evolution – Application Areas – The XMLHttpRequest Object and Protocol – difference between HTTP and XMLHttpRequest – Creating XMLHttpRequest object in javaScript – generating and customizing requests with XMLHttpRequest object – readystatechange event in XMLHttpRequest object – sending requests to server in synchronous and asynchronous methods – sending requests in get and post methods - Getting response from server using XMLHttpRequest – Dynamic and static responses - managing text response and XML response.

MODULE – IV: To understand jQuery

Introduction to jQuery – Need for jQuery – Problems with javaScript programming – advantages of using jQuery – jQuery syntax – integrating jQuery into a webpage – ready function – element selectors – events and binding method – HTML Dom Management – accessing element properties – changing element properties – managing element styles and classes – element dimensions - HTML DOM traversal – Selector filtering - jQuery effects – hide – slide – fade – toggle – stop - jQuery animations – callback functions – timer functions – AJAX using jQuery - load, get and post functions – jQuery form validation – jQuery multi-level navigation menu.

TEXT BOOK(S)

1. HTML, CSS and JavaScript All in One, Sams Teach Yourself: Covering HTML5, CSS3, and

jQuery Sams Publishing; 2 edition.

REFERENCE

1. HTML 5 Black Book: Covers CSS3, Javascript, XML, XHTML, AJAX, PHP and jQuery. By Kogent Learning Solutions Inc